IoT- Based Project— Road Traffic Flow Forecasting with LSTM



INVERTIS UNIVERSITY BAREILLY

BTech Deep Learning · Semester V · Section D
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Project: Road Traffic Flow Forecasting with LSTM
Mentor: Mr. Raushan Raj

Members Name

S.No.	Name	Student ID	Roll No.
1	Sachin	BCS2024275	2410303416
2	NIHARIKA TANDON	BCS2024313	2410303308
3	Anubhav Kumar	BCS2024314	2410303091
4	Aksa Iman	BCS2024316	2410303050
5	Pankaj Yadav	BCS2024317	2410303327
6	Priyanshu Yadav	BCS2024318	2410303363
7	ANSHIKA BHARDWAJ	BCS2024321	2410303087
8	Krish Rohila	BCS2024322	2410303237
9	AYUSH RATHORE	BCS2024326	2410303140
10	SMRITI KUMARI	BCS2024327	2410303486
11	MOHAMMAD AYAN	BCS2024329	2410303268
12	HEMANT RATHORE	BCS2024330	2410303204
13	SHREY	BCS2024331	2410303477
14	VISHAL BABU	BCS2024333	2410303557

INTRODUCTION

Introduction to IoT and Road Traffic Flow Forecasting

IoT (Internet of Things) means everyday objects with sensors that collect data and share it over a network. On roads, these devices turn traffic into a live record that forecasting can learn from.

Capture

Roadside sensors and counters record vehicle count and speed, and send the data to a central server.

Anticipate

Forecasting models read this history and estimate traffic flow for the next few minutes.

Act

Managers get the forecast early and can adjust signals or routes before a jam builds up.

PROBLEM STATEMENT

Traffic congestion and rising vehicle counts are outpacing manual management.

Congestion also adds delays and pollution, and manual control reacts too late.

01

Increasing vehicles

More vehicles mean bottlenecks keep coming back, causing long delays that are hard to predict.

02

Delays, pollution and accidents

Stop-and-go traffic wastes time and fuel, adds air pollution, and raises the risk of accidents.

03

Manual management

Officers watch junctions by hand and react late, so they cannot plan interventions in advance.

OBJECTIVES

Our objectives: forecast traffic flow and support earlier decisions.

Manual monitoring reacts too late, so we built a system that predicts traffic flow and supports earlier decisions.

Forecast

Predict short-term traffic flow using past and sensor data.

Reduce congestion

Give early forecasts that support decisions to reduce congestion.

Dashboard

Provide a simple dashboard that shows forecast and actual traffic flow.

Develop system

Build an IoT and LSTM system that runs from data input to prediction.

EXISTING VS PROPOSED SYSTEM

IoT, machine learning, and LSTM forecasting replace reactive monitoring.

Instead of only watching what is happening now, the system prepares for what comes next.

TRADITIONAL MONITORING

Traditional monitoring

We only see traffic as it is right now, so we react once a jam has already started. The picture is often patchy.

PROPOSED SYSTEM

IoT + ML + LSTM forecasting

Roadside IoT sensors feed traffic data to an LSTM model, which predicts upcoming flow so managers get warned early.

Monitoring tells us what is happening. Forecasting helps us decide what to do next.

TECHNOLOGIES USED

Six tools carry the project from data to dashboard.

Each tool has one job, from cleaning the data to showing the forecast.

Technology	Role in the project
Python	Main language for data processing, model building, and the pipeline.
Machine Learning	Learns patterns from past traffic records to support forecasting.
LSTM	Recurrent network that remembers earlier readings to predict the next flow.
Pandas	Loads, cleans, and organises the time-stamped traffic data.
NumPy	Handles the numerical arrays and sequence windows fed to the model.
Streamlit	Builds the interactive dashboard that shows forecasts to managers.

Dataset Description

The model learns from time-stamped traffic records, so the data must be clean, ordered, and scaled before training.

Data source

[insert dataset source and time period]

Features

- Timestamp of each reading
- Vehicle count and average speed
- Sensor or junction ID

Time-based data

Readings are recorded at regular intervals, so each row has a clear time order.

Preprocessing

- Missing values: [insert method used]
- Scaling: values scaled to a fixed range, for example 0–1
- Sorted by time, then split into train and test sets

Dataset overview: 48,204 records, 9 features, including weather fields, date_time and traffic_volume.

SYSTEM ARCHITECTURE & WORKFLOW

From data input to traffic output.

Six steps run in order, with an LSTM model that predicts the next traffic flow.

01

Data input

Collect traffic data from [insert dataset source].

02

Preprocessing

Clean, scale, and sort the readings by time.

03

Sequence creation

Cut the data into sliding windows for the LSTM.

04

LSTM training

Train the model on past sequences.

05

Prediction

Predict the next traffic flow.

06

Dashboard output

Show forecasts against actual flow in Streamlit.

Traffic output

The predicted flow is scaled back to real vehicle counts and shown as a forecast, so managers see upcoming congestion early.

MODEL DEVELOPMENT

An LSTM network learns from past traffic sequences.

LSTM is a type of recurrent neural network built to remember patterns across time.

Why LSTM

Traffic depends on earlier hours and days. LSTM memory cells keep that context, which a simple model cannot.

Training process

Sequences of past readings are fed in, and the network is trained on them in repeated passes.

Loss minimization

The loss measures the gap between predicted and actual flow. Training adjusts the weights to make this gap smaller.

Prediction mechanism

The trained model takes the latest window of readings and outputs the next traffic flow value.

RESULTS & PERFORMANCE EVALUATION

Actual vs predicted traffic flow, measured with error metrics and a baseline.

Error metrics, baseline comparison, and graph explanation sit together for easy reading.

MAE

321.09

Average size of the error

MSE

260271.67

Squared error, which punishes big misses

RMSE

510.17

Error in the same units as traffic flow

Baseline comparison

We compare the LSTM with a simple previous-value forecast.

Average traffic by hour

Traffic peaks in the afternoon and is lowest overnight.

DASHBOARD & APPLICATIONS

A Streamlit dashboard shows forecasts to traffic managers.

The interface turns model output into charts that are easy to read and act on.

Dashboard features

- Forecast display for the next time steps
- Actual vs predicted line charts
- Traffic flow summary for each junction

Practical applications

- Signal timing and congestion alerts
- Route planning for commuters

CONCLUSION

The model helps us act before congestion builds.

Forecasting gives traffic managers time to respond, instead of reacting after the jam has formed.

Project summary

An IoT-fed LSTM pipeline that forecasts short-term traffic flow from roadside sensor data.

Purpose achieved

Early forecasts give managers time to adjust signals and routes before a jam forms.

Benefits and limits

Earlier warnings help, but limits include sensor coverage and data quality.

FUTURE SCOPE

Future improvements to this traffic forecasting project

Real-time sensor integration

Stream live counts and speeds from roadside sensors into the model.

Weather and event data

Add rain, holidays and events as inputs that explain sudden changes in flow.

Multi-step forecasting

Predict several future time steps, not only the next one.

Better deep-learning models

Test stronger architectures against the current LSTM baseline.

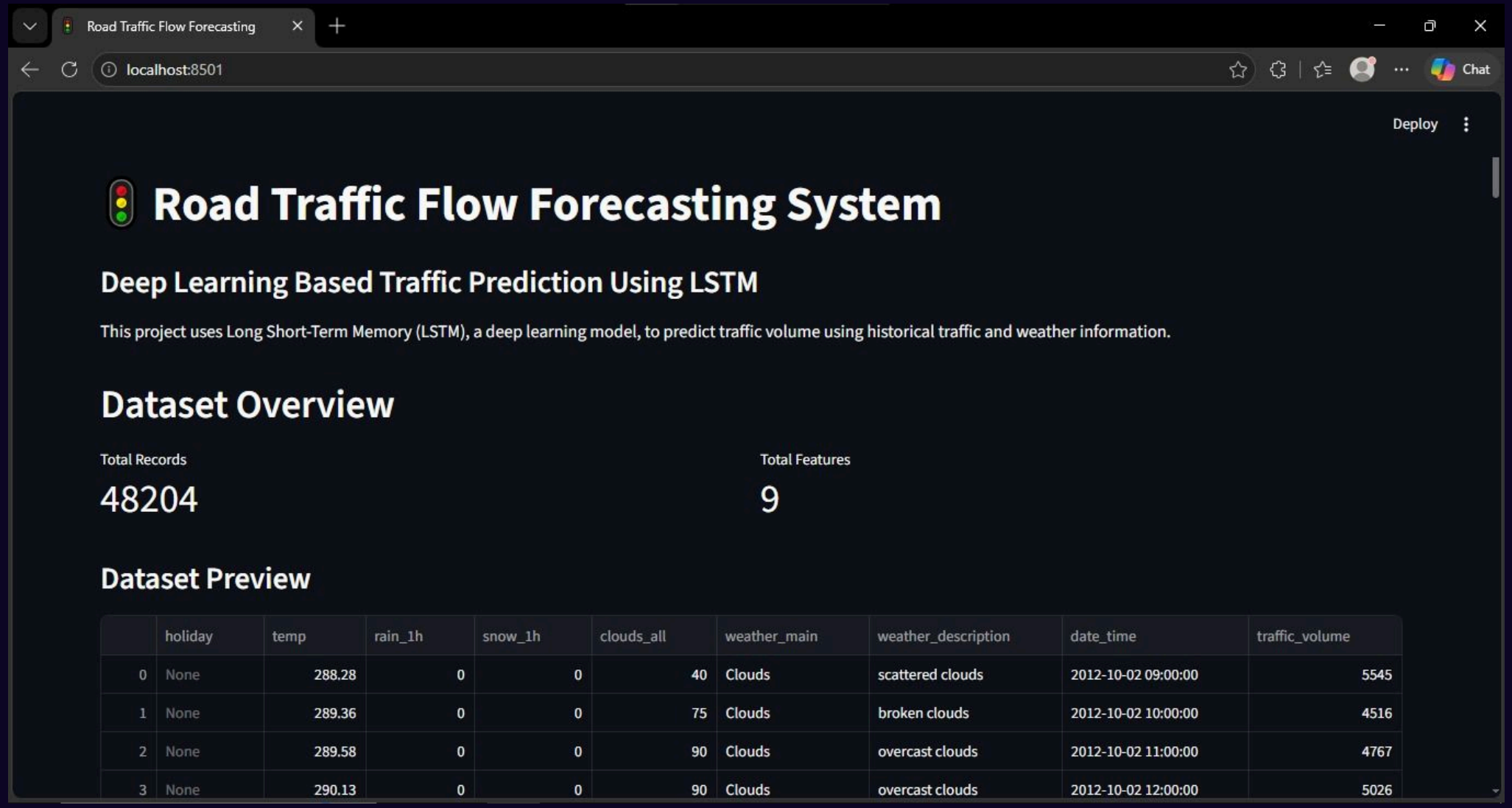
Edge and cloud deployment

Run inference near junctions and scale the system through the cloud.

Alert systems

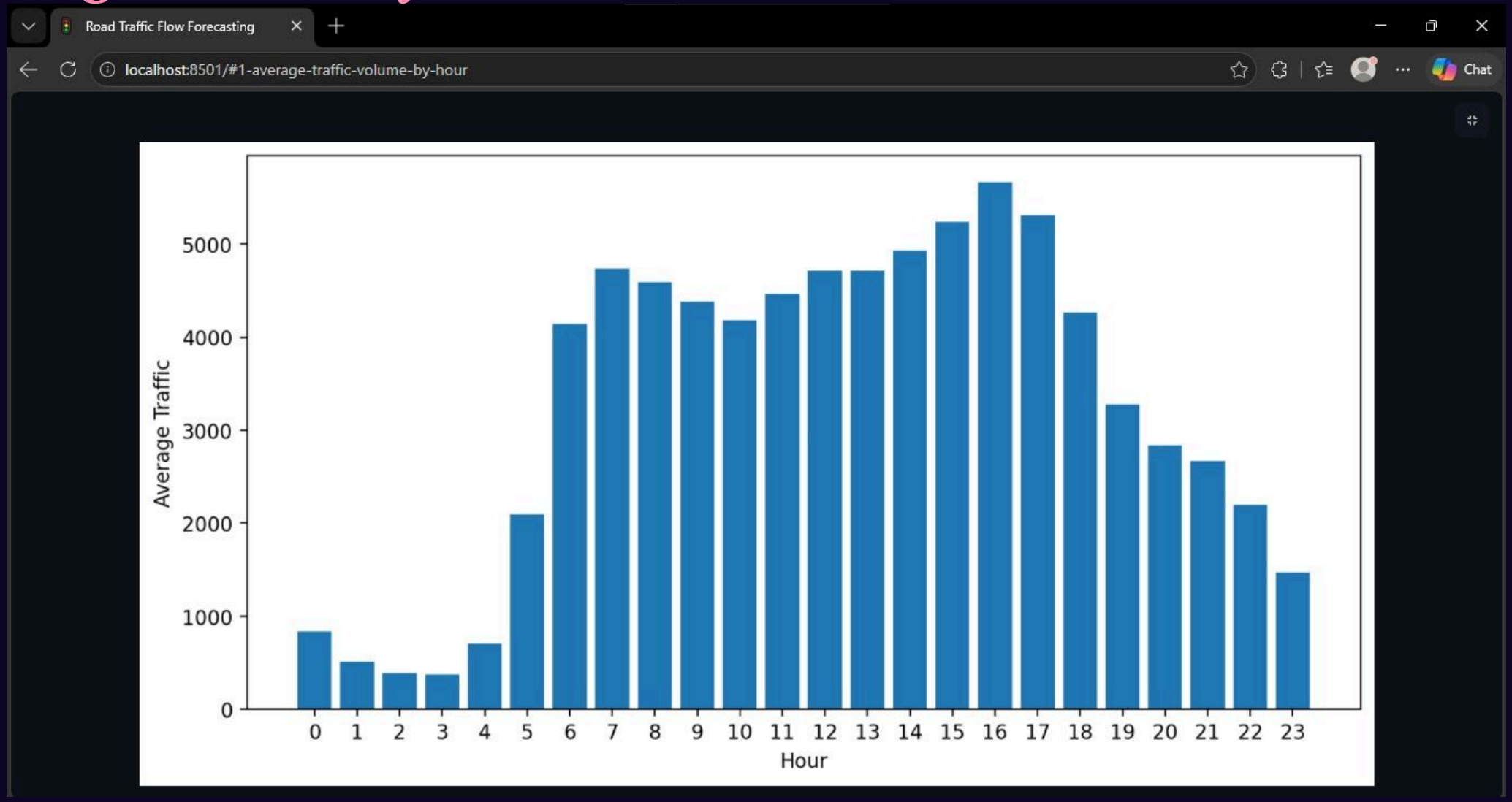
Send automatic congestion alerts to traffic managers and drivers.

Dataset Overview Dashboard

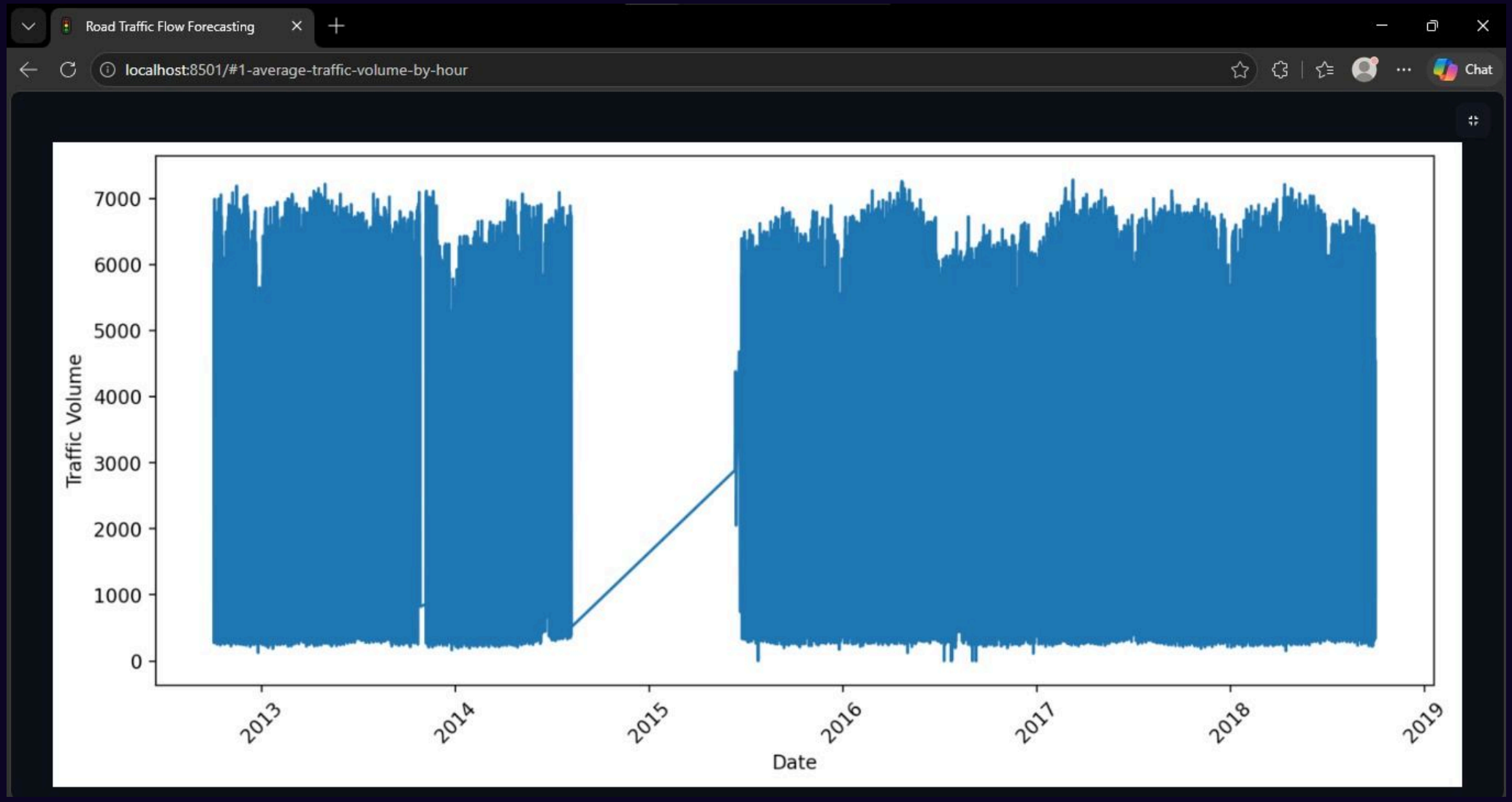


	holiday	temp	rain_1h	snow_1h	clouds_all	weather_main	weather_description	date_time	traffic_volume
0	None	288.28	0	0	40	Clouds	scattered clouds	2012-10-02 09:00:00	5545
1	None	289.36	0	0	75	Clouds	broken clouds	2012-10-02 10:00:00	4516
2	None	289.58	0	0	90	Clouds	overcast clouds	2012-10-02 11:00:00	4767
3	None	290.13	0	0	90	Clouds	overcast clouds	2012-10-02 12:00:00	5026

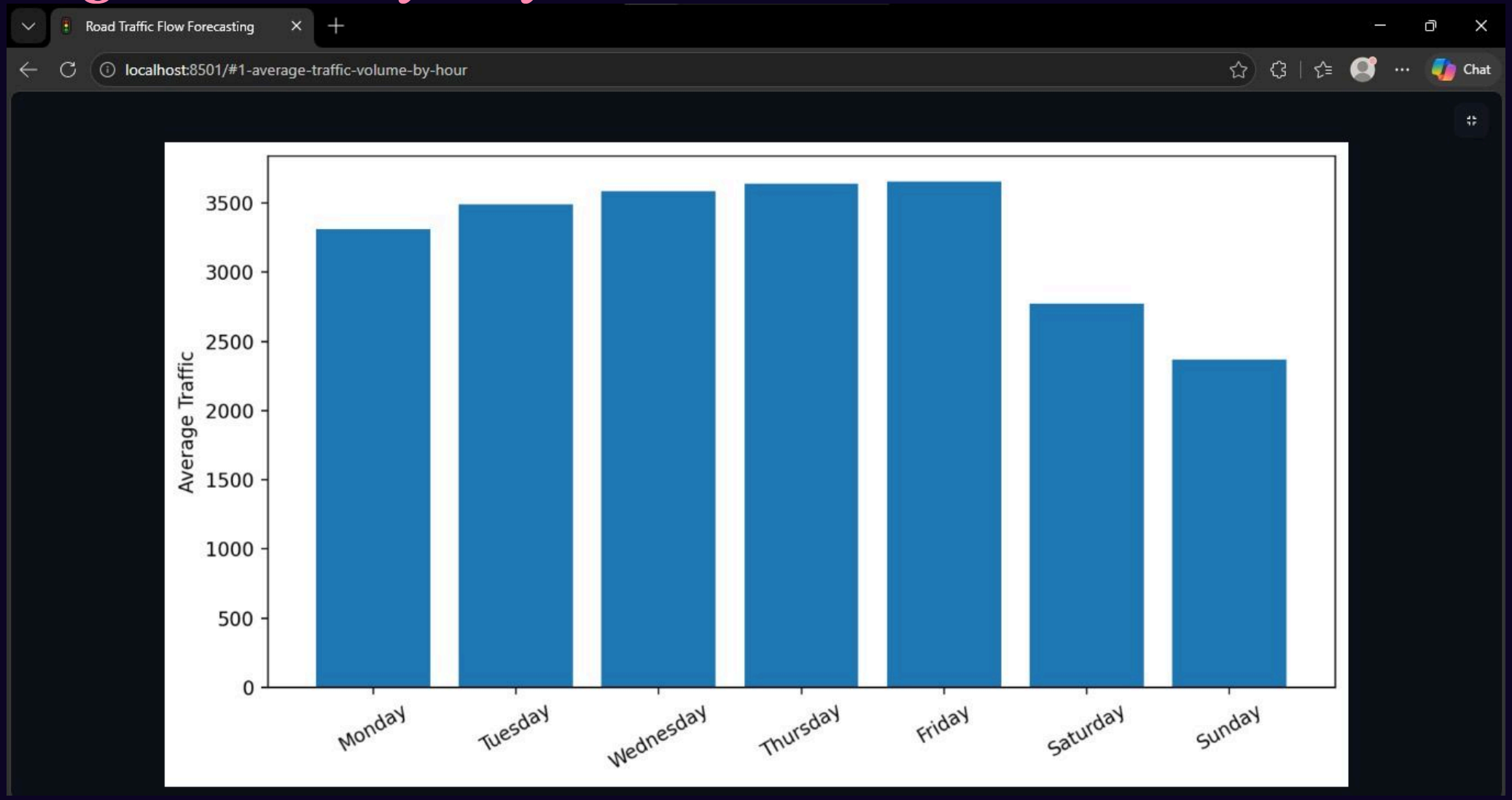
Average Traffic by Hour



Traffic Volume Over Time



Average Traffic by Day

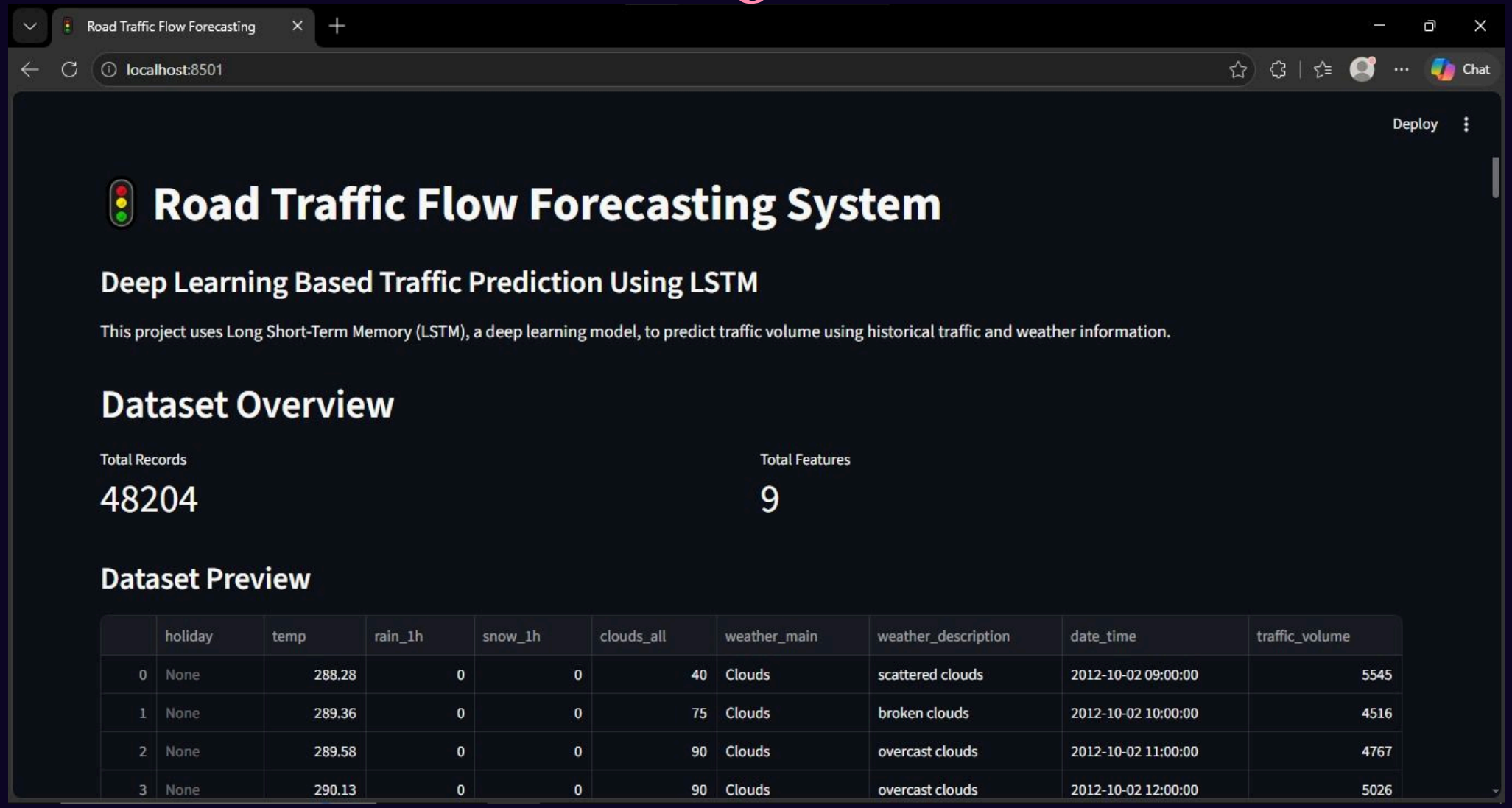


Model Performance Evaluation



	Metric	Baseline	LSTM
0	MAE	483.36	321.09
1	MSE	543168.22	260271.67
2	RMSE	737	510.17

Road Traffic Flow Forecasting Dashboard



	holiday	temp	rain_1h	snow_1h	clouds_all	weather_main	weather_description	date_time	traffic_volume
0	None	288.28	0	0	40	Clouds	scattered clouds	2012-10-02 09:00:00	5545
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Thank You